

Gravity as a Thermodynamic Derivative: The HIGM Informational Vacuum Perspective

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(This model was developed in the framework of theoretical physics and AI)

Abstract:

Modern cosmology faces a "Dark Sector" crisis, with the Hubble Tension and early-galaxy formation anomalies challenging the Λ CDM paradigm. This article proposes that gravity is not a fundamental force but an emergent entropic phenomenon. Using the Hamouda Informational Gravity Model (HIGM), we demonstrate that gravitational attraction is a manifestation of Informational Impedance within a data-dense vacuum. By treating the universe as an informational software system, we resolve the Hubble Tension ($H_0 \sim 73.5$ km/s/Mpc) and provide a physical mechanism for accelerated structural growth without the need for Dark Matter particles.

Keywords: *HIGM, Emergent Gravity, Informational Vacuum, Entropy Friction, Hubble Tension, Thermodynamics, Shannon Entropy.*

1. Introduction: The Software of Space

For a century, General Relativity has described gravity as the geometry of space. However, as we enter 2026, the failure to detect Dark Matter particles and the persistent 67 vs 73 km/s/Mpc discrepancy in the Hubble Constant suggest that geometry is not the foundation. The HIGM proposes that thermodynamics—specifically Information Entropy—is the true fundamental layer. In this view, space is not an empty stage but an Informational Vacuum that processes data carried by radiation.

2. The Thermodynamic Origin of Gravity

In the HIGM framework, gravity emerges from the Second Law of Thermodynamics.

- The Mechanism: Infrared Information Flux (IR flux) from stars and galaxies saturates the vacuum with data.
- The Force: As information density increases, an entropic gradient is formed. Matter is naturally "pushed" toward regions of higher entropy, creating the effect we perceive as gravitational pull.
- The 9.05% Boost: This thermodynamic pressure provides the additional velocity observed in galactic rotation curves, effectively replacing the "Dark Matter" hypothesis with a measurable Informational Impedance.

3. Connecting HIGM to Multi-Messenger Anomalies

The thermodynamic nature of gravity is most visible in its interaction with light and waves:

- ❖ Hubble Tension: The expansion rate is dynamic. As the universe's total entropy increases, the "informational push" accelerates the expansion, reconciling early-universe CMB data with modern observations.
- ❖ Entropy Friction: Gravitational waves are "entropic shockwaves." Unlike light, which defines the vacuum's speed, waves experience a ~5-microsecond latency (delay) due to the processing overhead of the Informational Vacuum.

4. Conclusion: A New Research Pillar

The HIGM model successfully bridges the gap between Quantum Information and General Relativity. By establishing thermodynamics as the primary driver of cosmic evolution, we move away from invisible matter toward a testable, informational universe. The validation of this model at the Gravity2026 workshop in Korea marks a transition from "Material Physics" to "Informational Physics," offering a stable roadmap for the next century of cosmology.

5. Selected References

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